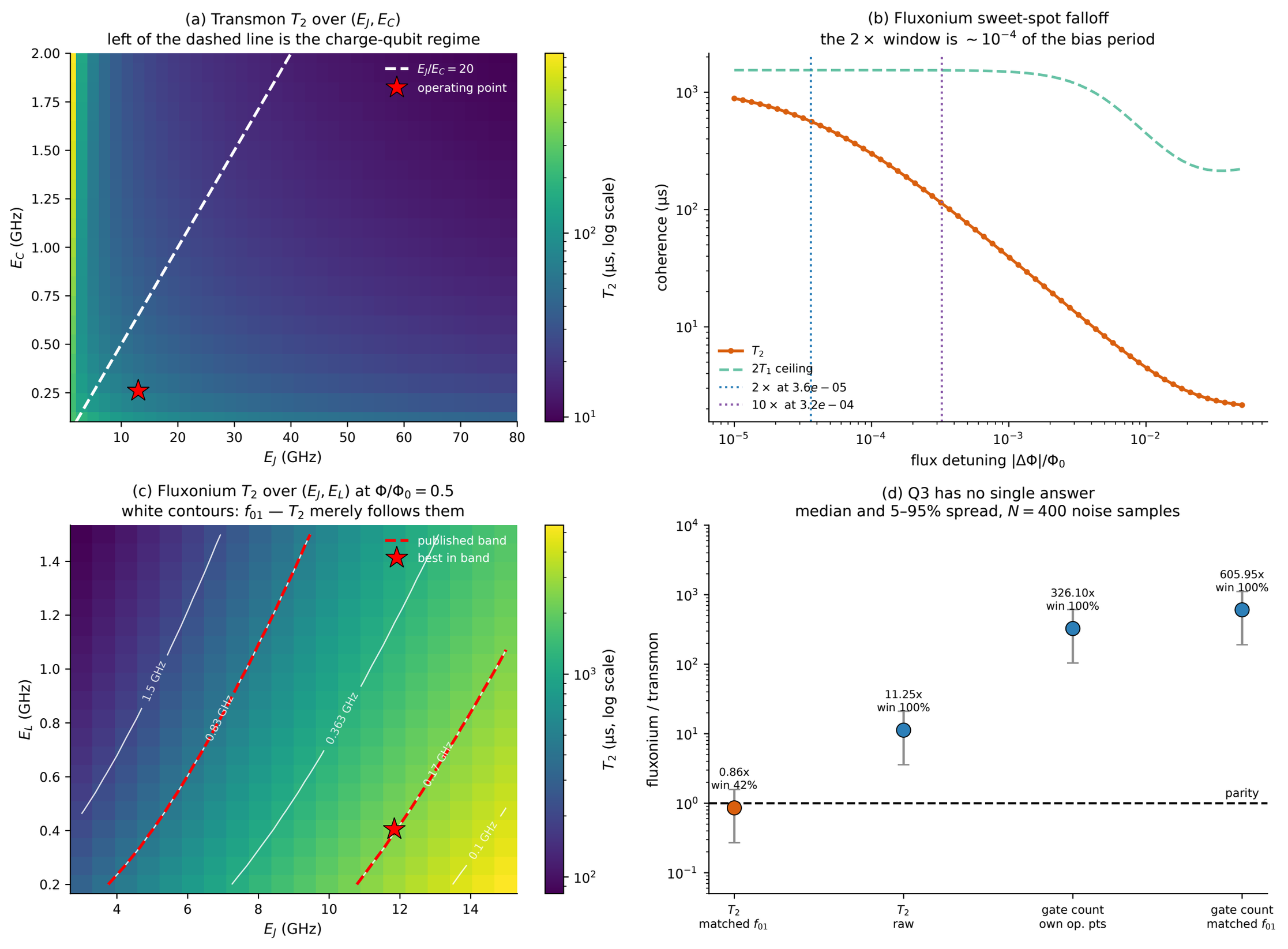

(a) Transmon T_2 over (E_J, E_C)
left of the dashed line is the charge-qubit regime
$E_J/E_C = 20$
operating point
E_C (GHz)
E_J (GHz)
T_2 (µs, log scale)
(b) Fluxonium sweet-spot falloff
the 2 × window is $\sim 10^{-4}$ of the bias period
coherence (µs)
T_2
$2T_1$ ceiling
2 × at 3.6e − 05
10 × at 3.2e − 04
flux detuning $|\Delta\Phi|/\Phi_0$
(c) Fluxonium T_2 over (E_J, E_L) at $\Phi/\Phi_0 = 0.5$
white contours: f_{01} — T_2 merely follows them
published band
best in band
1.5 GHz
0.83 GHz
0.363 GHz
0.17 GHz
0.1 GHz
E_L (GHz)
E_J (GHz)
T_2 (µs, log scale)
(d) Q3 has no single answer
median and 5–95% spread, $N = 400$ noise samples
fluxonium / transmon
0.86x
win 42%
11.25x
win 100%
326.10x
win 100%
605.95x
win 100%
parity
T_2
matched f_{01}
T_2
raw
gate count
own op. pts
gate count
matched f_{01}